

TW12: Data Visualization

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose
Create honest bar, line, histogram, scatter, stacked, and box plots that support a focused business story.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Create an honest category comparison

Scenario

A director wants to compare net revenue across regions and channels.

INSTRUCTIONS

1. Clean the necessary sales columns and calculate net revenue.
2. Create a sorted regional revenue bar chart with a zero baseline.
3. Create a horizontal channel revenue chart with direct labels.
4. Include titles, units, readable ordering, and a source/scope note.

REQUIRED SUBMISSION

- Two publication-ready bar charts.
- The exact aggregation tables used by the charts.
- A two-sentence finding and one limitation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Tell a time-series story

Scenario

Monthly sales contain trend, seasonality, and missing values.

INSTRUCTIONS

1. Parse and sort the monthly data.
2. Plot raw sales with missing values visible.
3. Add 3-month and 12-month trailing means.
4. Annotate the highest observed month without calling the rolling line a forecast.

REQUIRED SUBMISSION

- A chronological line chart with three clearly distinguished series.
- A table identifying the peak and missing periods.
- A brief interpretation discussing smoothing and lag.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Investigate a distribution with histograms

Scenario

The business wants to understand individual transaction values rather than only totals.

INSTRUCTIONS

1. Calculate clean net revenue per transaction.
2. Compare histograms using 6, 12, and 24 bins on shared scales.
3. Add mean and median reference lines.
4. Explain skew, unusual values, and the effect of bin choice.

REQUIRED SUBMISSION

- A three-panel histogram comparison.
- A descriptive-statistics table.
- A four-to-six sentence distribution interpretation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Visualise and quantify a relationship

Scenario

Management suspects marketing spend is related to transaction revenue.

INSTRUCTIONS

1. Create a scatter plot with transparency and suitable labels.
2. Colour points by channel without using more categories than the audience can read.
3. Add a fitted linear line and calculate Pearson correlation.
4. Inspect influential observations and explain why correlation does not establish causation.

REQUIRED SUBMISSION

- A labelled scatter plot with legend and fitted line.
- Correlation coefficient and p-value.
- An interpretation separating association, influence, and causation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Build a two-panel executive dashboard

Scenario

An executive needs one view of regional channel composition and one view of transaction-value distributions.

INSTRUCTIONS

1. Create a 100% stacked regional channel-revenue chart.
2. Verify each non-empty row sums to 1.
3. Create box plots of net revenue by channel with a numerical summary.
4. Write one headline, one recommendation, and two limitations.

REQUIRED SUBMISSION

- One coherent two-panel dashboard.
- The pivot, normalised share table, and distribution summary.
- Evidence-led text that does not hide absolute-size limitations.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____